Dashboards in Databricks

Turning raw data into decisions — a practical guide for data engineers, analytics engineers, and data-literate product managers.

DATABRICKS SQL

DATA VISUALISATION

Why Dashboards Matter

Data on its own has no inherent value — it only becomes valuable when it drives action. Dashboards serve as the critical bridge between raw data and meaningful business decisions, making insights accessible to everyone from engineers to executives.

📊 Data

Raw, unprocessed facts stored across tables, streams, and pipelines — the foundation of everything.

💡 Insights

Aggregated, contextualised views that reveal patterns, trends, and anomalies worth acting on.

✅ Decisions

Confident, timely actions taken by stakeholders who trust the data in front of them.

What Is a Dashboard?

A dashboard is a curated, visual interface that aggregates data from one or more sources into a single, coherent view. Rather than writing ad-hoc queries, consumers of data can monitor KPIs, explore trends, and answer questions at a glance — without needing to touch SQL themselves.

Visual Representation

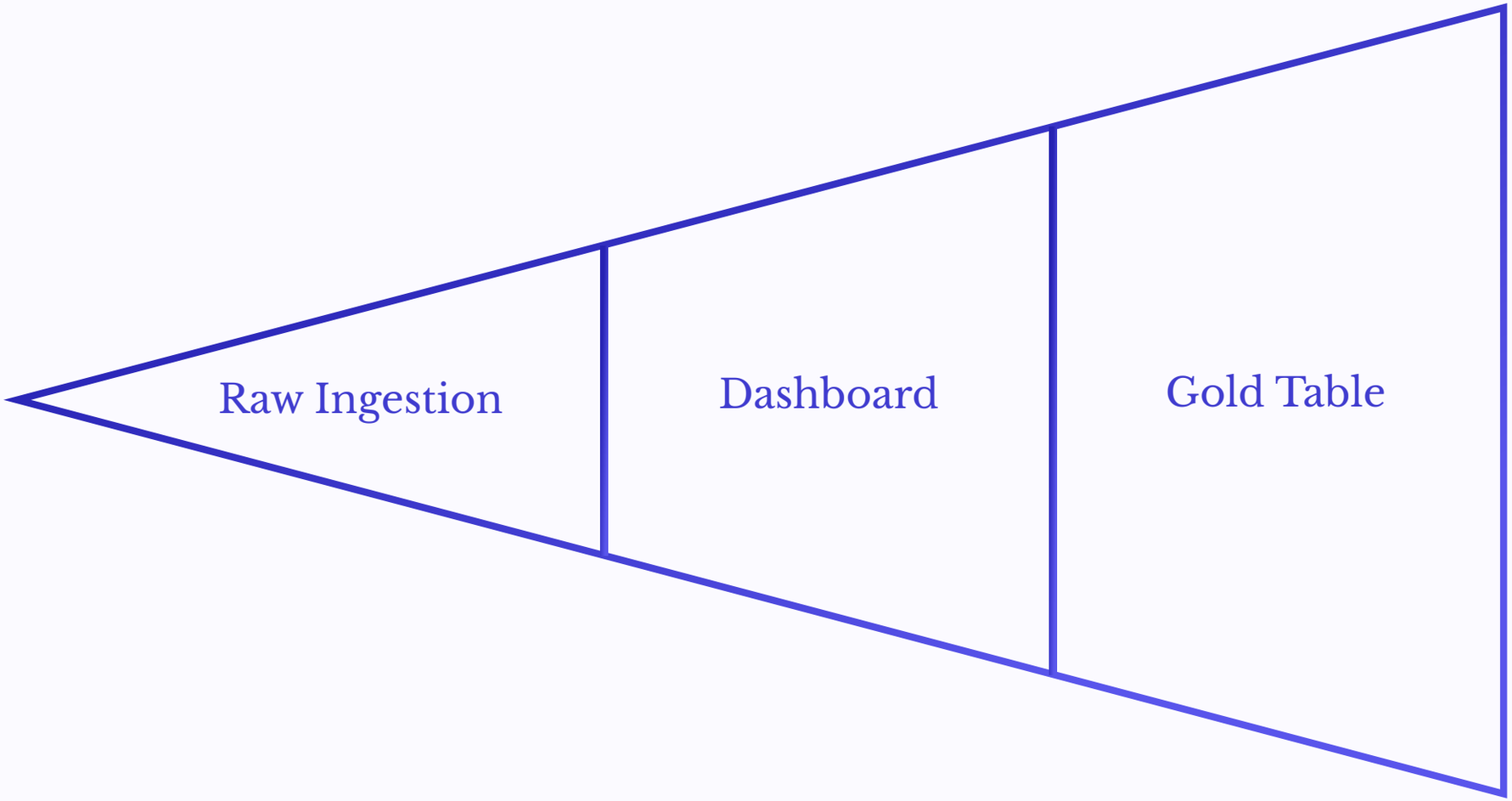
Charts, tables, and KPI tiles translate complex query results into immediately understandable visuals. The right visualisation makes the right insight obvious.

Aggregated Insights

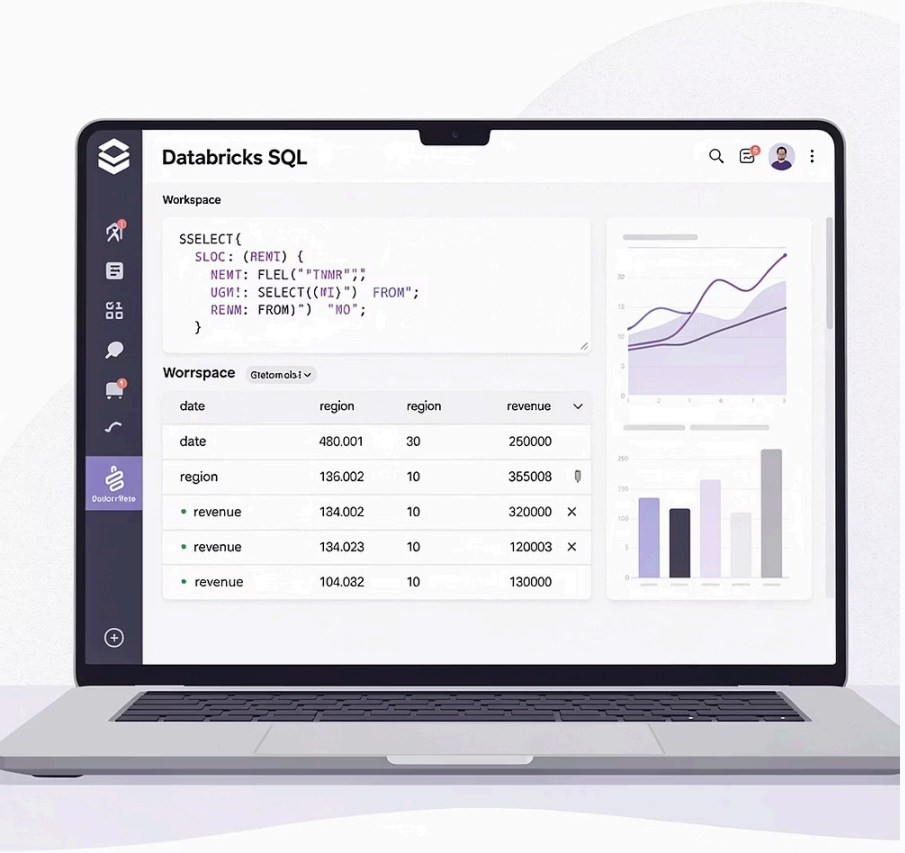
Dashboards combine multiple queries into a unified view — bringing together revenue metrics, pipeline health, user activity, and more on a single screen. Context is everything.

How Data Flows into a Dashboard

Every great dashboard starts long before a visualisation is rendered. Understanding the end-to-end flow helps you build dashboards that are reliable, performant, and trustworthy.



Following the medallion architecture ensures your dashboards are built on curated, high-quality Gold tables — not raw or intermediate data — keeping query performance fast and results consistent.



Databricks SQL Dashboards

Databricks SQL provides a purpose-built environment for creating dashboards directly within the Lakehouse. Queries run against Delta tables using standard SQL, and results are rendered as interactive visuals — all without leaving the platform.

Query-Based

Every visualisation is powered by a SQL query. You write the logic; Databricks handles execution against your cluster or SQL warehouse, returning results ready to visualise.

Interactive

Users can filter, drill down, and explore data dynamically using query parameters and widget controls — turning a static report into a live analytical tool.

Lakehouse-Native

Dashboards sit alongside your notebooks, jobs, and tables in Unity Catalog — governed, versioned, and secured within the same platform your engineers already use.

Building Your First Dashboard

Creating a dashboard in Databricks SQL follows a clear, repeatable workflow. Each step builds on the last, taking you from a blank canvas to a shareable, interactive view in minutes.

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Write Your Query

Open the SQL Editor and write a query against your Gold table. Use parameters (e.g. `{{date_range}}`) to make it dynamic and reusable across different dashboard contexts.

Create a Visualisation

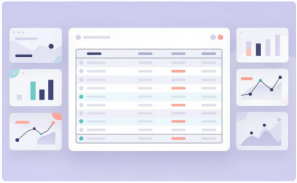
From the query result, choose a visualisation type — bar chart, line chart, table, or KPI tile. Configure axes, labels, and colours to make the insight immediately clear.

Add Widgets to the Dashboard

Add your visualisation as a widget to a new or existing dashboard. Arrange multiple widgets on the canvas, add text annotations, and use filter widgets to link query parameters across charts.

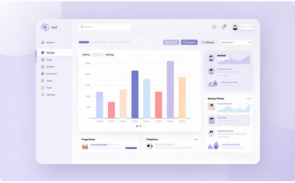
Types of Visualisations

Choosing the right visualisation is as important as writing the right query. Each chart type serves a distinct analytical purpose — use the one that makes the insight most obvious for your audience.



Table

Best for detailed, row-level data where precision matters. Ideal for operational reports, exception lists, and debugging pipeline outputs.



Bar Chart

Perfect for comparing discrete categories — revenue by region, events by type, or counts by status. Easy to read at a glance for any audience.

Line Chart

The go-to for time-series data. Reveals trends, seasonality, and anomalies over a continuous time axis — essential for monitoring pipelines and business metrics.

KPI Tile

A single, prominent number with optional trend indicator. Use for top-level metrics like total revenue, active users, or SLA compliance that executives need at a glance.

Scheduling & Alerts

A dashboard that requires manual refreshing is a dashboard that will eventually show stale data. Databricks SQL lets you automate both data refresh and proactive notification — so your team is always working from current numbers.

⏱️ Automatic Refresh

Schedule dashboards to re-run their underlying queries at a defined cadence — hourly, daily, or custom intervals. Results update automatically, keeping consumers confident without manual intervention.

- Set refresh frequency per dashboard
- Runs against your SQL warehouse on schedule
- No manual trigger needed

🔔 Threshold Alerts

Define alert conditions directly on query results. When a metric crosses a threshold — such as error rate above 5% or revenue below target — Databricks notifies your team via email or webhook integration.

- Alert on any numeric query result
- Integrates with Slack, PagerDuty, email
- Reduce time-to-detection on data issues

Sharing & Access Control

Dashboards are only useful if the right people can see them — and only safe if the wrong people cannot. Databricks SQL integrates with Unity Catalog to provide fine-grained, role-based access control across all dashboard assets.

Role-Based Access

Assign **View**, **Edit**, or **Manage** permissions to individual users or groups. Engineers can edit queries whilst stakeholders receive read-only access — no risk of accidental modification.

Secure Sharing

Share dashboards via a persistent URL within your Databricks workspace. Permissions are enforced at the platform level — recipients only see data their Unity Catalog role permits them to query.

Workspace Governance

All dashboard assets — queries, visualisations, and schedules — are managed within the Databricks workspace, ensuring auditability, lineage tracking, and compliance with your data governance policies.

A Pipeline Is Complete Only When Insights Are Consumed

> "You can have the cleanest Gold table, the most optimised pipeline, and the most robust ingestion layer — but if no one is reading the dashboard, the data has not yet done its job."

The final step in every data pipeline is not a dbt model or a Delta table — it is the moment a stakeholder reads a number, understands a trend, and makes a better decision. Build dashboards that close that loop.

Build on Gold

Always connect dashboards to curated, tested Gold tables.

Automate & Alert

Schedule refreshes and set thresholds so insights arrive proactively.

Share Securely

Govern access so the right people see the right data, always.